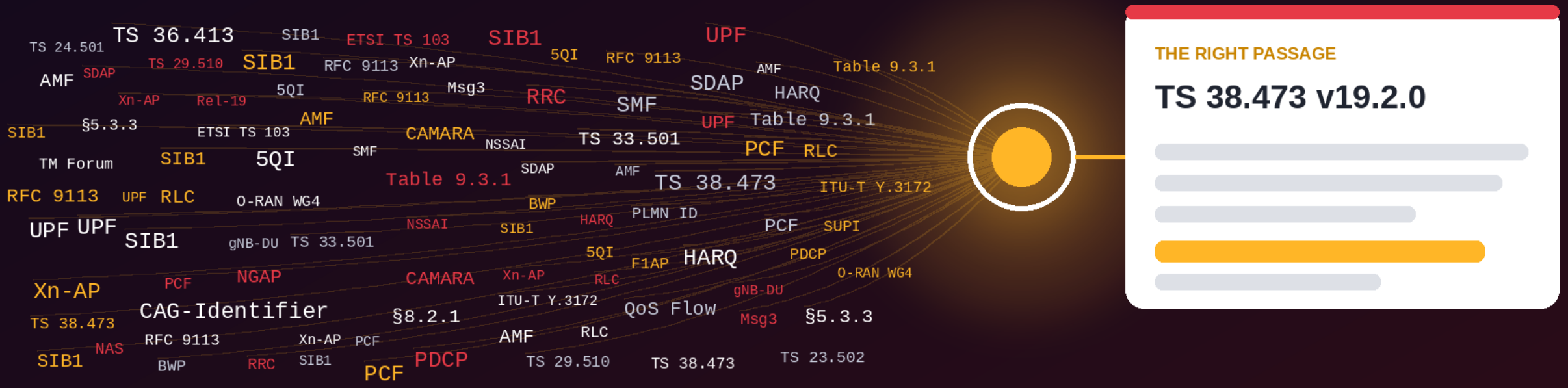
TELCORETRIEVE CHALLENGE

LLMs don’t speak 3GPP.

PRIZE POOL
€25,000
in total prizes

Liam Düro

UCL, Machine Learning Researcher



GSMA OPEN TELCO AI

TelcoRetrieve

One layer of telco knowledge every AI model can reason over it.

ONE LAYER

Telco knowledge in one retrieval layer

Open and vendor-neutral, so every model can reason over telecom standards.

CROSS-SDO RETRIEVAL

One index across many standards bodies

3GPP, ITU-T, IETF, IEEE, ETSI and more in one pipeline.

Fork the Pipeline to Start

Your task: make LLMs read 3GPP

TelcoRetrieve is the open retrieval layer. The challenge is to make it better.

THE TASK

1 track

Improve the pipeline

Change how context is retrieved and assembled, or fine-tune the models. Everyone works on the same 3GPP Rel-19 corpus.

HOW IT IS JUDGED

70 / 30

Accuracy and token cost

70% answer accuracy, 30% token efficiency. A leaner prompt scores higher. Minimum accuracy 80%.

WHAT YOU WIN

€25K

Prizes and MWC Barcelona

Cash for the top five, MWC Barcelona passes and travel for the top three. Strong ideas can join TelcoRetrieve.

13 NOV – 13 JAN 100 public questions from the start · 200 private questions on 8 Jan · top 10 go to code review

TELCORETRIEVE CHALLENGE · THE TRACK

One track: Pipeline and Model Optimization

3GPP corpus › **Chunking** › **Embedding** › **Index** › **Retrieval** › **Reranking** › **Generator** › Answer

Red steps are yours to change. Grey steps are fixed.

WHAT YOU CAN CHANGE

- **Chunking and index** — Strategy, size, overlap
- **Retrieval and reranking** — method, how many passages
- **The prompt** — how context is assembled
- **The models** — fine-tune the embedder or Gemma 3n

WHAT STAYS FIXED

3GPP Rel-19 only
the corpus is provided

OTel-Reranker-0.6B
the reranker model

4,096 tokens
maximum prompt length

SUBMIT your answers and input prompts · **TOP 10** hand over the whole pipeline to be checked

* Embedder: OTel-Embedding-0.6B (farbodtavakkoli/OTel-Embedding-0.6B) · Reranker: OTel-Reranker-0.6B (farbodtavakkoli/OTel-Reranker-0.6B) · Base model: Gemma 3n E4B instruct (google/gemma-3n-e4b-it)

What you receive and what you submit

YOU RECEIVE

Multiple choice | 3GPP Rel-19 | TS 38.473 v19.2.0

According to the specification, what is the maximum number of CAG-Identifiers that can be broadcast in a cell? (according to 3GPP Rel-19, TS 38.473 v19.2.0)

- **A** 16
- **B** 12
- **C** 10

also true / false

YOU SUBMIT

One CSV, header `ID,input_prompt,answer`

1	`ID`	Copied unchanged. Keep it as text.
2	`input_prompt`	What Gemma sees. Max 4,096 tokens.
3	`answer`	One letter A–E, or TRUE / FALSE.

```
"123456789012","You are an expert on 3GPP… Question: …","B"
```

HOW IT RUNS

13 NOV

100 public questions

Submit up to 300 times. The public leaderboard updates after every submission.

8 – 13 JAN

+200 private questions

Released before the close. Submit these too. Scores stay hidden and decide the top 10.

15 – 31 JAN

Code review

Top 10 hand over their code. Zindi reruns it, including a hidden evaluation set, then publishes final results.

How submissions are scored

PUBLIC LEADERBOARD

100 public questions Live from 13 Nov · scores shown after every submission

PRIVATE LEADERBOARD · HIDDEN UNTIL THE END

200 private questions Released 8 Jan · scores revealed only after the close

ACCURACY

Multiple-choice and true/false

Exact match with the gold answer. One option letter or TRUE / FALSE only; anything else scores 0.

Accuracy = 100 × correct / questions

CODE

`accuracy.py`

prints accuracy, 0–100

TOKEN EFFICIENCY

Tokens in your input_prompt

Counted over your input_prompts. A prompt over 4,096 tokens rejects the whole submission.

Token efficiency = 100 × B / (B + T)

B = 850 × questions; T = your tokens

CODE

`token_efficiency.py`

tiktoken cl100k_base

FINAL SCORE

One mixed metric

Fewer tokens raise the score; wrong answers lower it.

Score out of 100. Minimum accuracy: 80%

0.7 × Accuracy + 0.3 × Token efficiency

CODE

`overall_score.py`

prints the final score, 0–100

LEADERBOARD **Public ranking: the 100 public questions. Top 10: chosen on the 200 private questions. Final ranking: after the code review.**

TOP 10 **Code review: hand over your whole pipeline (code and configuration). Zindi reruns it, including a hidden evaluation set.**

Scoring, a worked example

Same 100 questions, three pipelines. Baseline B = 850 × 100 = 85,000 tokens. Illustrative numbers.

PIPELINE A

Retrieve a lot

Accuracy	**90%**
Tokens (T)	**85,000**
	850 tokens per question
Token efficiency	**50.0**

FINAL SCORE
78.0

PIPELINE B

Trim the prompt

Accuracy	**85%**
Tokens (T)	**50,000**
	500 tokens per question
Token efficiency	**63.0**

FINAL SCORE
78.4

PIPELINE C

One tight chunk

Accuracy	**82%**
Tokens (T)	**25,000**
	250 tokens per question
Token efficiency	**77.3**

FINAL SCORE
80.6

Accuracy counts for 70%, token efficiency for 30%. A leaner prompt can beat a more accurate pipeline: C wins with 8 points less accuracy than A.

RULES Minimum accuracy 80% · every input_prompt at most 4,096 tokens (one longer prompt rejects the whole submission) · tokens counted with tiktoken cl100k_base

What not to do

DON'T Leave rows out

```
99 rows for 100 questions
```

Result `Missing entries for IDs …`

Do One row for every question ID in the file.

DON'T Repeat an ID

```
"09436ccaeebf",… (two rows)
```

Result `Duplicate IDs …`

Do Each ID exactly once.

DON'T Send an empty prompt

```
"09436ccaeebf","","B"
```

Result `Empty input_prompt for IDs …`

Do The full prompt Gemma saw, never blank.

DON'T Go over 4,096 tokens

```
input_prompt = 4,500 tokens
```

Result `input_prompt over 4096 tokens for IDs …`

The whole submission is rejected.

Do Trim passages. Check with token_efficiency.py.

DON'T Answer in sentences

```
"The answer is B" "b) 12"
```

Result Cannot be read, so it scores 0.

Do B or TRUE / FALSE. Nothing else.

DON'T Let Excel touch the IDs

```
123456789012 → 1.23457E+11
```

Result `Missing entries for IDs …`

Do Never open or save the CSV in Excel. Read IDs as text.

ALSO Header exactly `ID,input_prompt,answer`: no extra columns, no missing header.

CHECK FIRST Run accuracy.py, token_efficiency.py and overall_score.py on your own CSV

Your ideas can join TelcoRetrieve

We review the top 10 solutions that reach the second phase and select some of their best contributions for potential integration into the TelcoRetrieve pipeline.

NOVELTY

Original ideas

Introduces original approaches, techniques or architectural improvements that go beyond standard RAG optimisation and offer clear potential to advance the TelcoRetrieve pipeline.

Performance

Demonstrable gains in retrieval accuracy and token efficiency.

Generalisation

Performs well on unseen and private test data, not just the public leaderboard.

Robustness

Handles ambiguous queries, missing information and different telecom domains reliably.

Efficiency

Reasonable inference latency, GPU/CPU requirements and computational cost.

Technical quality

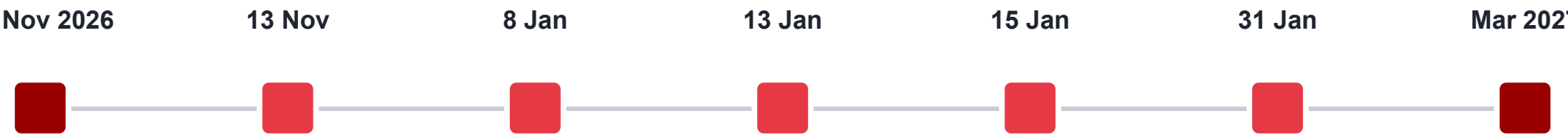
Clean, modular, maintainable code that integrates with the existing pipeline.

Reproducibility

Code runs reliably, with clear setup instructions and documented dependencies.

TELCORETRIEVE CHALLENGE · 2026 — 2027

From announcement to awards



Announcement

Challenge announced at MWC Doha

Competition opens

Toolkit and 100 public questions. Public leaderboard goes live

200 new questions

Private questions released. Submit these too

Competition closes

Last day to submit. Top 10 selected on the 200 private questions

Code review

Top 10 hand over their code. Zindi reruns it, including a hidden evaluation set

Final results

Final ranking published after the code review

Awards

Awards at MWC Barcelona

PUBLIC LEADERBOARD

100 questions, live from 13 Nov

PRIVATE LEADERBOARD

200 questions from 8 Jan, hidden until the end

TELCORETRIEVE CHALLENGE · RESOURCES

Everything you need to build

Toolkit Znbne/telco-retrieve
The RAG pipeline itself: ingest, search, rerank and answer from the command line.
GITHUB

Indexed data
Ready-made Rel-19 chunks and index plus the question data. Start tuning without re-ingesting.
HUGGING FACE

Guidebook Znbne/telcoretrieve-cookbook
Worked examples for each stage: chunking, retrieval, reranking and generation.
GITHUB

Raw dataset GSMA/3GPP
The 3GPP specifications as text, if you want to build your own chunks.
HUGGING FACE

Training data telecom-standards-corpus
Telecom standards text for fine-tuning your embedder or Gemma 3n.
HUGGING FACE

Scoring scripts Znbne/telco-retrieve
accuracy.py, token_efficiency.py and overall_score.py: check your CSV before you submit.
GITHUB

Prizes and where to start

PRIZE POOL: €25K

PRIZES

1st	€8,000
2nd	€5,000
3rd	€3,500
4th	€2,000
5th	€1,500

+ MWC Barcelona Leader Pass (worth €2,000)
+ up to €1,500 in costs to attend MWC Barcelona in March 2027 (one representative)

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Plus 10,000 Zindi points available.

START HERE

github.com/Znbne/telco-retrieve

Competition opens 13 Nov 2026. Closes 13 Jan 2027.



MWC BARCELONA

Winners are announced at MWC Barcelona, March 2027

Join the challenge

Search: TelcoRetrieve Challenge on Zindi

zindi.world/competitions/telcoretrieve-challenge-unlocking-telco-knowledge

Registration closes 6 Jan 2027 · teams of up to 3